

Supplementary Material: Localizing to Debias: A Patch-Level Benchmark and Baseline for Weakly Supervised Spatial Anomaly Detection

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GitHub Code

S1 Dataset Annotation

The released annotations cover both training and test splits, creating 11,855 tubelets across 2,541 annotated events across UCF-Crime, MSAD, and XD-Violence. These annotations can serve as a supervision resource for future fully supervised or semi-supervised localization research.

S1.1 Frame-level Consistency

As noted in Section 3, existing spatial annotations for anomaly localization are available only for a subset of UCF-Crime [2]. These annotations suffer from inconsistencies, such as variable bounding-box sizes across frames and noisy localization of the same event. Figure S1 illustrates these issues. In the first two rows, the annotated boxes inconsistently capture the anomaly region, sometimes including unnecessary background or missing relevant areas. In contrast, our annotations are designed to be spatially relaxed, including the natural motion of objects without excessive frame-to-frame shifts. The lower examples show that the prior annotations often omit contextual elements and object interactions; for instance, enclosing only a single actor can misrepresent an event as normal when viewed out of context, as in the robbery event in the third row.



Fig. S1: Comparison of the spatial annotations produced in UCFCrime2Local [2] (red) and our annotation (light blue).

S1.2 Benchmark Statistics

To characterize the spatial and temporal properties of the WSVAL annotations, we report a set of validation metrics computed from the bounding-box tracks across all three datasets. These metrics quantify how much of the frame the annotated anomaly region occupies, how densely anomaly activity is distributed within each event, and how the temporal structure of individual tubelets varies across datasets. Together they serve two purposes: validating annotation completeness and consistency, and contextualizing dataset-level differences that bear on evaluation behavior. Table S1 reports overall and per-category statistics.

Background Ratio. Across all datasets, background constitutes the dominant fraction of the frame (0.77, 0.90, and 0.57 for UCF-Crime, MSAD, and XD-Violence respectively), with individual categories such as Shooting and People Falling in MSAD reaching fractions above 0.96. This confirms the structural condition motivating WSVAL: under video-level weak supervision, the background is spatially co-activated with the anomaly signal throughout training, and spatial annotations are necessary to audit whether model predictions are grounded in the anomaly region or driven by background shortcuts.

Tubelet Duration. Tubelet duration varies substantially across datasets (mean 382, 69, and 70 frames for UCF-Crime, MSAD, and XD-Violence respectively) and is informative beyond reflecting event length alone. Longer tubelets require a model to sustain spatially coherent predictions over extended periods without reverting to background cues, such that the smoothness and consistency of predictions across the tubelet becomes a meaningful signal of reasoning quality. At the same time, datasets with higher tubelet-per-event counts, e.g. arising from scene cuts or multi-camera transitions, carry a structural evaluation risk: noisy spatial predictions can accumulate true positives across many short independent segments, inflating PAUC and PAP relative to what would be observed on longer, continuous anomaly tracks.

S2 Extended Methods

S2.1 Detailed DTFE Formulation

We provide here the detailed formulation of the Dynamic Tubelet Feature Encoder, including the context-mixing, token sampling, and masked attention operations sketched conceptually in Section 4.2. Formally, we represent a video segment as a collection of N tubelets t_i , where each is embedded by a Conv3D into a feature tensor x_i

$$X = \{x_1, x_2, \dots, x_N\}, \quad x_i \in \mathbb{R}^{(t_z \times T \times D)}, \quad (1)$$

where t_z , $T = H_P \times W_P$ (with $H_P=H/P$, $W_P=W/P$), and D are the split size, the number of spatial tubelet locations per sub-segment (flattened from the

Table S1: WSVAD annotation statistics per dataset. *T/Ev*: tubelets per event; *BBox Ratio*: mean fraction of frame area covered by the anomaly bbox; *Bkg Ratio*: complementary background fraction; *Ano Frac*: fraction of event frames covered by at least one tubelet; *Tube Dur*: mean tubelet duration in frames. Values report mean $\pm$ std.

Dataset	Category	Events	Tubelets	T/Ev	BBox Ratio	Bkg Ratio	Ano Frac	Tube Dur (fr)
UCF-Crime	Abuse	93	374	4.02 $\pm$ 4.26	0.220 $\pm$ 0.126	0.780 $\pm$ 0.126	0.948 $\pm$ 0.114	179 $\pm$ 259
	Arrest	51	284	5.57 $\pm$ 4.32	0.244 $\pm$ 0.168	0.756 $\pm$ 0.168	0.952 $\pm$ 0.100	566 $\pm$ 1027
	Arson	59	121	2.05 $\pm$ 1.97	0.250 $\pm$ 0.148	0.750 $\pm$ 0.148	0.940 $\pm$ 0.109	683 $\pm$ 2272
	Assault	57	143	2.51 $\pm$ 2.17	0.278 $\pm$ 0.161	0.722 $\pm$ 0.161	0.932 $\pm$ 0.126	304 $\pm$ 527
	Burglary	110	210	1.91 $\pm$ 1.65	0.278 $\pm$ 0.177	0.722 $\pm$ 0.177	0.901 $\pm$ 0.191	1143 $\pm$ 3428
	Explosion	22	23	1.05 $\pm$ 0.21	0.221 $\pm$ 0.134	0.780 $\pm$ 0.134	1.000 $\pm$ 0.000	348 $\pm$ 386
	Fighting	59	147	2.49 $\pm$ 2.22	0.284 $\pm$ 0.162	0.716 $\pm$ 0.162	0.931 $\pm$ 0.124	487 $\pm$ 896
	RoadAccidents	159	179	1.13 $\pm$ 0.57	0.204 $\pm$ 0.157	0.796 $\pm$ 0.157	0.965 $\pm$ 0.086	124 $\pm$ 257
	Robbery	149	508	3.41 $\pm$ 3.30	0.312 $\pm$ 0.175	0.688 $\pm$ 0.175	0.926 $\pm$ 0.135	389 $\pm$ 723
	Shooting	60	294	4.90 $\pm$ 3.69	0.156 $\pm$ 0.103	0.844 $\pm$ 0.103	0.850 $\pm$ 0.212	102 $\pm$ 172
	Shoplifting	67	155	2.31 $\pm$ 2.08	0.226 $\pm$ 0.153	0.774 $\pm$ 0.153	0.855 $\pm$ 0.215	239 $\pm$ 455
	Stealing	112	489	4.37 $\pm$ 3.86	0.159 $\pm$ 0.107	0.841 $\pm$ 0.107	0.908 $\pm$ 0.166	381 $\pm$ 701
	Vandalism	63	183	2.91 $\pm$ 3.12	0.200 $\pm$ 0.130	0.800 $\pm$ 0.130	0.852 $\pm$ 0.209	222 $\pm$ 378
	<i>Overall</i>	<i>1061</i>	<i>3110</i>	<i>2.93$\pm$3.41</i>	<i>0.230$\pm$0.163</i>	<i>0.770$\pm$0.163</i>	<i>0.920$\pm$0.183</i>	<i>382$\pm$1317</i>
MSAD	Assault	15	57	3.80 $\pm$ 2.43	0.077 $\pm$ 0.055	0.923 $\pm$ 0.055	1.000 $\pm$ 0.000	84 $\pm$ 116
	Explosion	15	34	2.27 $\pm$ 1.39	0.186 $\pm$ 0.139	0.814 $\pm$ 0.139	1.000 $\pm$ 0.000	107 $\pm$ 140
	Fighting	15	71	4.73 $\pm$ 3.37	0.133 $\pm$ 0.092	0.867 $\pm$ 0.092	0.938 $\pm$ 0.133	89 $\pm$ 107
	Fire	25	56	2.24 $\pm$ 1.51	0.105 $\pm$ 0.083	0.895 $\pm$ 0.083	0.983 $\pm$ 0.049	320 $\pm$ 688
	Object Falling	22	124	5.64 $\pm$ 3.62	0.118 $\pm$ 0.075	0.882 $\pm$ 0.075	0.892 $\pm$ 0.196	33 $\pm$ 53
	People Falling	44	175	3.98 $\pm$ 2.77	0.063 $\pm$ 0.046	0.937 $\pm$ 0.046	0.839 $\pm$ 0.243	24 $\pm$ 37
	Robbery	30	251	8.37 $\pm$ 4.69	0.072 $\pm$ 0.055	0.928 $\pm$ 0.055	0.956 $\pm$ 0.118	65 $\pm$ 79
	Shooting	17	137	8.06 $\pm$ 5.47	0.040 $\pm$ 0.034	0.960 $\pm$ 0.034	0.979 $\pm$ 0.063	43 $\pm$ 55
	Traffic Accident	39	185	4.74 $\pm$ 3.67	0.140 $\pm$ 0.102	0.860 $\pm$ 0.102	0.906 $\pm$ 0.176	32 $\pm$ 38
	Vandalism	14	24	1.71 $\pm$ 1.07	0.116 $\pm$ 0.087	0.884 $\pm$ 0.087	0.938 $\pm$ 0.134	144 $\pm$ 194
	Water Incident	10	26	2.60 $\pm$ 1.78	0.210 $\pm$ 0.136	0.790 $\pm$ 0.136	1.000 $\pm$ 0.000	222 $\pm$ 311
	<i>Overall</i>	<i>246</i>	<i>1140</i>	<i>4.63$\pm$4.20</i>	<i>0.096$\pm$0.103</i>	<i>0.904$\pm$0.103</i>	<i>0.931$\pm$0.169</i>	<i>69$\pm$174</i>
XD-Violence	Abuse	12	78	6.50 $\pm$ 5.17	0.411 $\pm$ 0.260	0.589 $\pm$ 0.260	0.880 $\pm$ 0.175	40 $\pm$ 58
	Car Accident	471	1914	4.06 $\pm$ 3.73	0.200 $\pm$ 0.229	0.800 $\pm$ 0.229	0.981 $\pm$ 0.083	31 $\pm$ 64
	Explosion	143	525	3.67 $\pm$ 5.82	0.432 $\pm$ 0.265	0.568 $\pm$ 0.265	0.992 $\pm$ 0.071	82 $\pm$ 201
	Fighting	247	2073	8.39 $\pm$ 10.54	0.512 $\pm$ 0.230	0.488 $\pm$ 0.230	0.972 $\pm$ 0.095	60 $\pm$ 124
	Riot	157	1452	9.25 $\pm$ 12.25	0.632 $\pm$ 0.241	0.368 $\pm$ 0.241	0.989 $\pm$ 0.038	187 $\pm$ 333
	Shooting	204	1563	7.66 $\pm$ 8.71	0.406 $\pm$ 0.220	0.594 $\pm$ 0.220	0.958 $\pm$ 0.120	23 $\pm$ 32
	<i>Overall</i>	<i>1234</i>	<i>7605</i>	<i>6.16$\pm$8.27</i>	<i>0.428$\pm$0.278</i>	<i>0.572$\pm$0.278</i>	<i>0.977$\pm$0.090</i>	<i>70$\pm$182</i>

$H_P \times W_P$ grid), and the feature dimension, respectively. Then, the spatial and temporal positions are encoded with a learnable positional map p_i

$$\tilde{x}_i = x_i + p_i, \quad P = \{p_1, \dots, p_N\}, \quad (2)$$

yielding a position-aware sequence. To aggregate information from all spatial, temporal, and spatio-temporal dimensions, we adopt a spatio-temporal context mixing strategy inspired by prior token-mixing and dynamic sparsification works [1, 5]. Essentially, the input token $\tilde{x}_i$ is projected to the different dimensions, before aggregated into a mix of spatial, temporal, and spatio-temporal information. This is achieved by applying the input token to multiple MLP layers, where each is a composition of a linear layer and a GELU activation. The spatial component is obtained as

$$\tilde{x}_i^{spatial} = \text{MLP}_s(\tilde{x}_i) \in \mathbb{R}^{(t_z \times T \times F_c)}, \quad (3)$$

where the feature size F_c is $\frac{D}{3}$. Similarly, temporal features are obtained by another MLP_s , followed by pooling along the temporal dimension and expanding the result:

$$\tilde{x}_i^{temp} = \text{AvgPool}(\text{MLP}_t(\tilde{x}_i), \hat{D}) \in \mathbb{R}^{(t_z \times 1 \times F_c)}. \quad (4)$$

Lastly, the spatio-temporal component is found by

$$\tilde{x}_i^{sptemp} = \text{AvgPool}(\text{MLP}_{spt}(\tilde{x}_i), \hat{D}) \in \mathbb{R}^{(1 \times 1 \times F_c)}. \quad (5)$$

The features are then aggregated along the last dimension forming the evidence of each token

$$\tilde{x}_i^{comix} = \text{Concat}(\tilde{x}_i^{temp}, \tilde{x}_i^{spatio}, \tilde{x}_i^{sptemp}), \quad (6)$$

where $\tilde{x}_i^{comix} \in \mathbb{R}^{(t_z \times T \times D)}$. The aggregated features are passed through a lightweight binary classifier $\phi(\cdot)$, implemented as three MLP layers with GELU activations. The classifier serves as the learned mechanism that identifies which spatio-temporal regions are anomaly-relevant vs. background noise, by producing per-token keep/drop probabilities. Formally,

$$\mathbf{a} = \text{Softmax}(\text{MLP}(\tilde{x}^{comix})) \in \mathbb{R}^{N \times 2}, \quad (7)$$

where $\mathbf{a}_{i,1}$ and $\mathbf{a}_{i,0}$ denote the keep and drop probabilities for the i -th token, respectively. Next, we prune the VAD-irrelevant tokens by a dynamic transformer [5], which progressively discards tokens that do not contribute to the final anomaly prediction. The pruning acts as discrete sampling of tokens, which prevents pruned tokens from contributing to the gradients during backpropagation. Since the regular computation of the self-attention is not feasible under this setting, we apply masked self-attention [3, 5], in which the interactions are restricted to the subset of active tubelets only. Given query, key, and value projections $Q, K, V \in \mathbb{R}^{N \times d}$, the masked attention is

$$A_{ij} = \frac{\exp(Q_i K_j^\top / \sqrt{d}) \cdot \mathcal{G}_{ij}}{\sum_{k=1}^N \exp(Q_i K_k^\top / \sqrt{d}) \cdot \mathcal{G}_{ik}}, \quad (8)$$

where $\mathcal{G} \in \{0, 1\}^{N \times N}$ is a binary mask ensuring that pruned tubelets do not contribute to interactions. The suppression of the irrelevant tubelets is performed by sampling a binary keep decision for each token using the Gumbel-Softmax trick

$$g_i \sim \text{GumbelSoftmax}(\mathbf{a}_i), \quad g_i \in \{0, 1\}, \quad (9)$$

where $g_i = 1$ indicates that token i is retained. The sampled decisions $\mathbf{g} \in \{0, 1\}^N$ are lifted to the pairwise attention mask $\mathcal{G} \in \{0, 1\}^{N \times N}$:

$$\mathcal{G}_{ij} = \begin{cases} 1, & i = j, \\ g_j, & i \neq j, \end{cases} \quad (10)$$

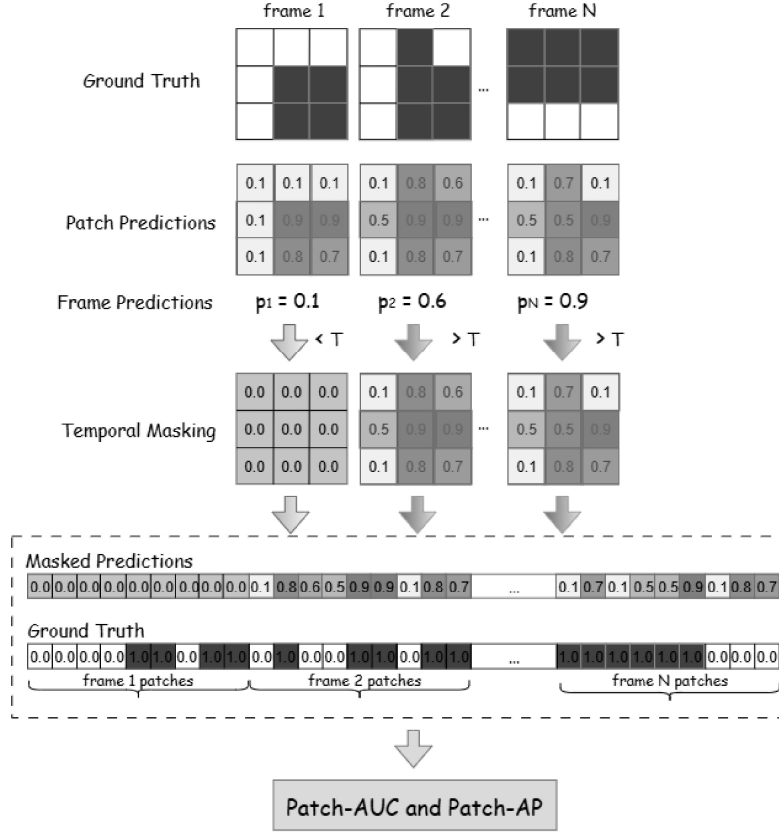


Fig. S2: The proposed patch-wise AUC and AP evaluation protocol.

where the indices i and j refer to the row and column positions in the attention mask applied to the self-attention matrix explained earlier. The mask is updated progressively across pruning layers:

$$\hat{\mathcal{G}} \leftarrow \hat{\mathcal{G}} \odot \mathcal{G}, \quad (11)$$

where $\odot$ is an element-wise matrix multiplication. The surviving tokens are then linearly mapped into an anomaly classification space to localize anomalies spatially. The localization is carried out by aggregating the responses of the retained tokens, such that tokens corresponding to discriminative tubelets produce higher anomaly scores. This allows the patch branch to operate as a weakly-supervised anomaly detector spatially.

S3 Evaluation Metrics

S3.1 Patch-level Metrics

Figure S2 demonstrates the proposed evaluation protocol. We compute AUC and AP at the patch level, where each frame is divided into 64 patches of 16×16 resolution. The patch scores of all frames are flattened and weighted by the corresponding temporal scores before calculating PAUC and PAP.

To obtain ground-truth patch labels, we measure the overlap between patches and the annotated anomaly bounding boxes. A patch is labeled positive if it overlaps at least 25% with any anomaly box in that frame. This threshold mitigates harsh penalties from bounding-box shifts between consecutive frames, and ensures that small-object regions are captured. For stricter evaluation, the overlap threshold can be increased to 50% or 100%, trading coverage for precision. We present the effect of different patch-box overlap thresholds in Section S5.2.

S3.2 Heatmap Generation

To visually evaluate the model, we produce heatmaps highlighting the most critical anomaly-related regions based on the top-20 patch outliers (top-10 and bottom-10). This choice is motivated by the fact that even the best models still exhibit relatively low precision, making patch-level classification challenging. However, highlighting these outliers demonstrates the capability of the proposed end-to-end framework to identify the most critical regions correctly, despite some misclassifications.

To generate the heatmaps, the model first produces tubelet predictions. Since the split size is 2 and the segment length is 16, the total number of tubelets in this setting is 128. Consequently, two predictions are obtained for each spatial location over the 16 frames: one for the first sub-segment and one for the second. These predictions are combined by selecting the highest score for each spatial location within the 16-frame segment. Based on these scores, heatmaps are generated using a soft Gaussian splatting technique. Specifically, patch coordinates and their anomaly scores are projected onto a low-resolution grid, where each patch contributes a Gaussian-weighted response proportional to its score. The resulting grid is then upsampled, smoothed, and overlaid on the video frame.

S4 Additional Experiments

S4.1 Spatio-Temporal Class-wise Performance

Tables S2–S4 present the class-wise performance of SST-WSVADL. Temporally, the table compares the per-class performance gains with the state-of-the-art temporal models [4, 8]. Clearly, SST-WSVADL significantly improves the class-level AUC and AP scores and maintains a balanced performance across classes compared to prior methods [4, 8]. Achieving a performance comparable to the

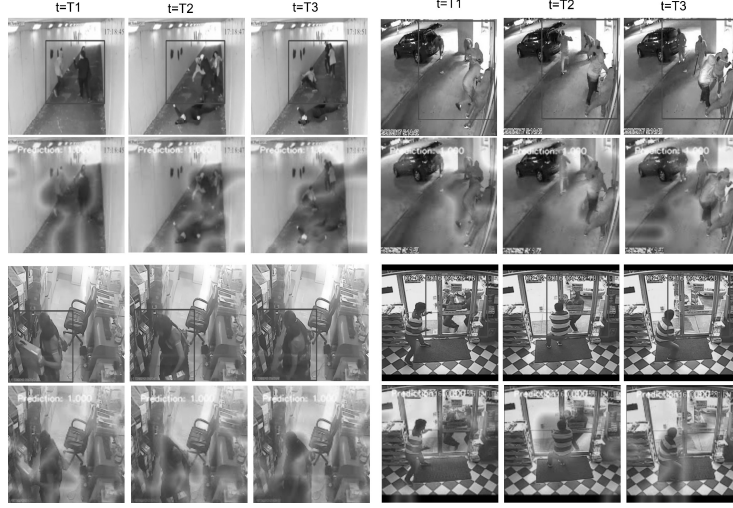


Fig. S3: Additional qualitative results from UCF-Crime (Burglary079, Fighting033, Robbery048, Shoplifting016).

multi-modal π -VAD model demonstrates that the joint spatio-temporal training may achieve the benefits of multi-modal training while avoiding its added complexity, as shown in the following section. Spatially, we analyse the patch separation capability (PAUC) and the spatial coverage (TIoU). We see that the classes with high temporal AUC performance (e.g. Abuse, Arrest, Arson, Burglary, and Stealing) also show a superior spatial PAUC, which indicates that the separation ability is aligned temporally and spatially. The TIoU metrics reflect that the model’s spatial coverage is only superior in cases of compact and large motion (e.g. Arrest, Assault, Burglary, Fighting, Explosion, Water-incident, and Riot). Other classes, such as Shoplifting, Abuse, Shooting, and Traffic accidents, show low TIoU which shows that the model is still weak when motion is slow or when multi-instance activities are present.

We further report patch-level metrics evaluated on anomaly frames only in Table S5, to set a reference for future works.

S4.2 Computational Complexity

Table S6 compares SST-WSVADL with representative weakly-supervised WSVAD methods in terms of parameter count and inference GFLOPs. FLOPs are reported for the task-specific detection modules, excluding backbone feature extraction for fair comparison across methods employing different feature encoders. For SST-WSVADL, this includes both the snippet-level and patch-level branches, including DTFE and PSA.

Although SST-WSVADL has a larger parameter count due to its dual-branch architecture, its computational cost remains moderate (12.39 GFLOPs). It is

Table S2: Class-wise temporal and spatial comparison on UCF-Crime.

Category	Temporal		Spatial	
	Class-wise AUC		PAUC	TIoU
	UR-DMU [8]	π -VAD [4]	SST-WSVADL (Ours)	
Abuse	81.00	80.00	96.00	82.27
Arrest	64.00	69.00	71.41	74.73
Arson	60.00	70.00	71.48	66.55
Assault	89.00	88.00	81.39	78.49
Burglary	77.00	79.00	83.29	82.10
Explosion	47.00	78.00	57.00	60.03
Fighting	79.00	91.00	89.00	80.80
RoadAccidents	68.00	91.00	64.67	63.13
Robbery	83.00	82.00	81.21	72.17
Shooting	77.00	86.00	82.48	82.05
Shoplifting	66.00	74.00	59.07	57.83
Stealing	85.00	90.00	92.28	84.97
Vandalism	90.00	92.00	86.26	84.84

Table S3: Class-wise temporal and spatial comparison on MSAD.

Category	Temporal		Spatial	
	Class-wise AUC		PAUC	TIoU
	UR-DMU [8]	π -VAD [4]	SST-WSVADL (Ours)	
Assault	50.66	70.00	75.48	71.00
Explosion	63.59	78.00	87.84	77.64
Water Incident	98.72	99.00	97.33	85.25
Fire	73.39	48.00	88.40	80.09
Object Falling	92.35	90.00	88.93	71.96
People Falling	43.13	48.00	59.41	64.10
Robbery	60.35	70.00	72.86	61.51
Vandalism	88.25	84.00	84.78	80.71
Traffic Accidents	67.74	71.00	76.74	70.30
Shooting	75.98	85.00	85.84	68.12
Fighting	82.84	84.00	88.50	78.62

Table S4: Class-wise temporal AP comparison on XD-Violence.

Category	Temporal		Spatial	
	Class-wise AP		PAUC	TIoU
	UR-DMU [8]	π -VAD [4]	SST-WSVADL (Ours)	
Fighting	58.00	90.00	85.78	84.91
Shooting	61.00	63.00	70.06	77.19
Riot	97.00	97.00	97.42	78.34
Abuse	62.00	73.00	72.81	84.71
Car Accident	34.00	40.00	52.16	65.77
Explosion	58.00	62.00	65.82	79.14

more efficient than prompt-based or multimodal frameworks, such as STPrompt [7] (44.8 GFLOPs) and π -VAD [4] (19.88 GFLOPs), while providing explicit spatial localization. Compared to UR-DMU, which performs temporal detection

Table S5: PAUC and PAP metrics measured for full and anomaly-only frames in the test set.

Dataset	PAUC	PAUC _{sub}	PAP	PAP _{sub}
UCF-Crime	88.45	58.76	25.80	65.96
MSAD	88.40	59.19	37.46	54.78
XD-Violence	81.00	61.94	68.20	82.43

Table S6: Computational complexity comparison.

Method	Parameters (M)↓	GFLOPS ↓
RTFM [6]	28	186.9
UR-DMU [8]	6.16	1.54
π -VAD [4]	82.81	19.88
STPrompt [7]	31.5	44.8
SST-WSVADL (Ours)	122.03	12.12

only (1.54 GFLOPs), SST-WSVADL introduces spatial reasoning at a controlled computational increase.

Pruning also affects computational cost, as shown in Table S7. However, the learned pruning rate varies across models, thereby the actual reduction in computation and memory depends on the dataset and model configuration. In the best performing model, the effect is minimal since the model learns to retain most tubelets. Thus, *task-wise*, the pruning strategy primarily benefits the learning and anomaly localization quality, while *cost-wise* it has little impact on efficiency.

It is worth noting that the pruning behaviour is also influenced by patch granularity. At smaller patch sizes, each tubelet covers a finer spatial region and therefore carries a relatively smaller portion of scene information. This makes background tubelets more separable from anomaly-relevant ones, possibly leading the model to higher rates of pruning and cost-level benefits of sparsification. This interaction between patch size and pruning rate is a direction for further exploration in future work.

Table S7: The learned token keeping rates of SST-WSVAD at each pruning layer, and the computational impact under different training settings on UCF-Crime.

Mode	Modules			Layer Keep Rate			Computational Impact GFLOPs
	DTFE	PSA	$\mathcal{L}_{motion}$	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_3$	
w. pruning	✓	✗	✗	0.72	0.45	0.23	7.60
	✓	✓	✗	0.46	0.44	0.44	7.56
	✓	✓	✓	0.94	0.94	0.93	12.12
w/o. pruning	✗	✓	✗	1.0	1.0	1.0	12.39

S5 Ablation Studies

S5.1 Effect of Motion Loss

Figure S4 illustrates the qualitative impact of motion-aware regularization on the sparsification process. Without motion loss, the model highlights scattered and spatially inconsistent patches, frequently activating static background regions. The resulting heatmaps exhibit multiple diffuse blobs, indicating that the pruning mechanism lacks a structural bias toward temporally dynamic evidence. When motion loss is incorporated, the highlighted regions become more spatially coherent and better aligned with the ground-truth anomalous areas (red boxes). The model consistently focuses on dynamically evolving entities, such as moving vehicles, interacting individuals, or flames, while suppressing activations on static or irrelevant regions (scene borders or background structures). This behavior demonstrates that motion loss regularizes tubelet selection by promoting motion-consistent tubelets during dynamic sparsification.

Table S8: The percentage of frames across the number of patches covering anomaly region in UCF-Crime anomalous test.

Dataset	Patch coverage	Frames percentage (%)
UCF-Crime	19/64	24.5%
	38/64	6.17%
	64/64	0.87%
MSAD	19/64	47.34%
	38/64	22.06%
	64/64	4.43%
XD-Violence	19/64	91.01%
	38/64	78.34%
	64/64	42.92%

S5.2 Patch-Box Overlap

Table S9 reports the spatial localization metrics under different patch-box overlap thresholds used to generate ground-truth patch labels. As the overlap threshold increases, PAP and TIoU decrease due to the stricter labeling criterion. However, both metrics remain relatively stable between 50% and 100% overlap, indicating that the localization performance does not collapse under stricter supervision. In contrast, PAUC remains nearly unchanged, showing that the model’s ranking ability is largely unaffected by the choice of overlap threshold. On XD-Violence, the higher overlapping thresholds show no changes in metrics, which is due to the large portion of anomalous frames that have bounding box equal to frame size, e.g. anomaly coverage equals 64 patches (Table S8).

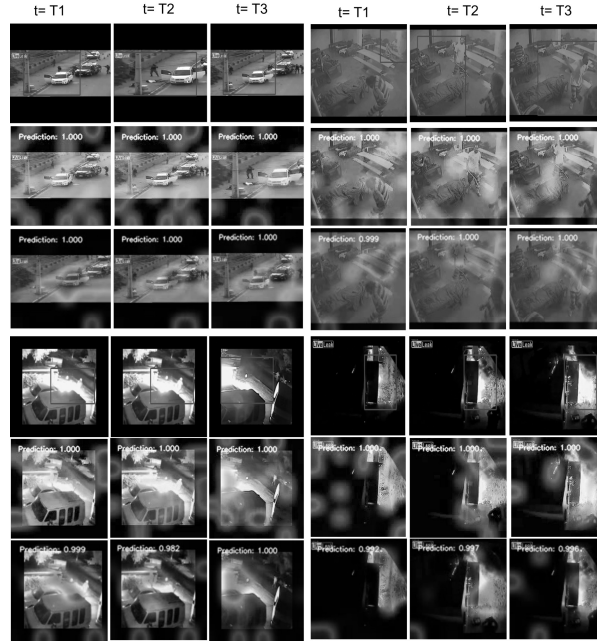


Fig. S4: Visual comparison of the patch outliers with and without motion loss in the patch branch objective. In each example, the top row shows the ground truth, the middle row depicts results without motion loss, and the bottom row shows results with motion loss applied.

Table S9: Spatial localization metrics evaluated under different overlap thresholds used to generate ground-truth patch labels.

Dataset	PAUC	PAP	TIoU	TIoU _{sub}
<i>25 % Overlap</i>				
UCF-Crime	88.45	25.80	14.59	26.25
MSAD	88.40	37.46	26.27	N/A
XD-Violence	81.00	68.20	29.49	N/A
<i>50% Overlap</i>				
UCF-Crime	88.49	24.23	11.50	21.07
MSAD	88.60	35.37	23.04	N/A
XD-Violence	80.43	65.70	28.23	N/A
<i>75% Overlap</i>				
UCF-Crime	88.69	22.73	9.98	16.28
MSAD	88.48	32.15	20.37	N/A
XD-Violence	79.96	63.36	27.05	N/A
<i>100% Overlap</i>				
UCF-Crime	88.41	20.10	8.37	11.88
MSAD	88.89	31.00	17.97	N/A
XD-Violence	79.96	63.36	27.05	N/A

S6 Limitations

Despite its strong performance, SST-WSVADL has several limitations. First, the patch-level branch (P-WSVAD) may still assign high anomaly scores to tubelets

in normal events. This behavior stems from the coarse MIL-based training approach, where supervision is provided at the video level and both normal and abnormal segments contribute to optimization. As a result, background motion or visually salient but non-anomalous dynamics can sometimes be amplified.

Second, spatial localization exhibits temporal inconsistency across consecutive snippets. For instance, given a sequence of five 16-frame snippets, the model may produce stable localization for a subset of them, while generating noisy or partially misaligned heatmaps in others. This reflects the inherent difficulty of enforcing fine-grained temporal coherence under weak supervision without explicit frame-level labels.

Addressing these limitations may require incorporating stronger temporal smoothness constraints, or conditioning motion-aware regularization on anomaly likelihood to prevent background dynamics in normal segments from influencing sparsification.

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